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Implementing ocular treatment with pencil beam scanning: the FHCC experience

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Abstract

Background The Fred Hutchinson Cancer Center (FHCC) has developed a novel in-house method for ocular proton therapy by adapting a pencil beam scanning (PBS) beamline and using a commercial treatment planning system. This manuscript outlines the workflow from simulation to treatment delivery and presents our experiences with the initial 40 patients.

Methods Key innovations of our treatment approach include in-house developed treatment chair and gaze localization systems, CT imaging for planning, and a Monte Carlo algorithm for dose calculation. We gathered data on patient characteristics, dose volume statistics, and total treatment time. Additionally, we examined the distances patients traveled to access ocular proton therapy. An example illustrating our treatment technique is also presented.

Results The average patient age at the time of treatment was 63.9 years. Tumor apical height ranged from 0.7 to 11.9 mm, and the largest basal diameter from 2.8 to 15.5 mm. GTV volumes ranged from 0.02 to 0.89 cc, while PTV volumes ranged from 0.17 to 2.27 cc. The D99% dose to GTV ranged from 5041 to 5242 cGy (RBE). The median mean dose to the lacrimal gland was 1570 cGy (RBE), while the median D2% doses to the optic nerve, macula, and optic disc were 4188 cGy (RBE), 5024 cGy (RBE), and 5133 cGy (RBE), respectively. Out of the 40 patients, 16 successfully met all treatment planning goals. The remaining patients did not meet some goals due to the target either abutting or being close (< 2 mm) laterally or distally to the OARs. The total treatment duration per fraction was approximately 25 min. Nearly one-third of the patients traveled around 900 miles to receive ocular treatment.

Conclusions The approach at FHCC demonstrates that ocular proton therapy can be effectively delivered using a general-purpose PBS beamline, providing a solution for centers without dedicated ocular beamlines.

Keywords Ocular proton therapy, Pencil beam scanning, General purpose beamline, Non-dedicated beamline, Uveal melanoma

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Background

With 3–5% of all melanoma cases [1, 2], ocular melanomas represent the most common intraocular malignancy in adults. Treatment options include enucleation, plaque brachytherapy, stereotactic radiotherapy, and proton beam therapy [3]. Proton therapy, introduced for ocular tumors in 1975 with pioneering work at Harvard [4], has become a vital modality due to its ability to deliver highly conformal doses while sparing surrounding critical ocular structures, primarily performed on dedicated beamlines. Unlike brachytherapy, which can be limited by tumor size and proximity to the optic disc, proton therapy remains effective for larger or unfavorably located tumors. Long-term studies report excellent clinical outcomes with proton therapy, including 5-year local control rates exceeding 90%, eye retention rates over 90%, and favorable vision preservation in selected patients [5–9]. These outcomes highlight not only the efficacy of proton therapy in tumor control—achievable by enucleation—but also its advantages in preserving vision and cosmesis, which are often the primary goals in treating ocular melanoma. Most proton therapy centers dedicated to ocular treatments are based on single-scattering beamlines with significant in-house developments to enable these treatments ([10, 11]). These centers typically do not perform planning on 3D images, instead relying on model-based solutions such as Eyeplan [11], EOPP (Varian Medical Systems) [11] and Octopus [11]. While the beam properties, such as lateral penumbra and distal fall-off, are generally favorable for dedicated beamlines, the dose calculation algorithms employed are often simplified, generally based on lookup tables. Replicating these legacy systems is challenging without significant investment in dedicated beamlines and antiquated treatment planning systems that have limited support and minimal development. As a result, many new proton therapy centers opt not to offer ocular treatments. Consequently, ocular proton therapy remains limited to only a few centers in the United States.

At the Fred Hutchinson Cancer Center (FHCC), we have developed an innovative approach to deliver ocular proton therapy by modifying a commercial general-purpose beamline and using a commercial treatment planning system. This approach enables full 3D treatment planning on the patient's head, employing Monte Carlo (MC) dose calculation algorithms and robust optimization. Patient doses are calculated and visualized on CT images, and dose-volume histograms (DVHs) are evaluated. Treatments are performed in a fixed beam room with the patient in a seated position, though this solution could also be implemented with a 360° gantry for supine treatments. We are sharing our initial experience of treating the first 40 patients, along with details of our workflow including immobilization, gaze localization,

treatment planning, patient-specific quality assurance (QA), and treatment delivery. Additionally, a treatment planning case study of a patient treated at our center is presented.

This work builds on earlier studies we conducted regarding beamline modifications [12], seated-position treatments [13], marker validation [14] and the use of a general-purpose commercial system [15] for treatment planning. We were able to achieve lateral penumbras in the range of 1.8 to 2.4 mm, depending on the beam range, which are similar to reported values [12] on the dedicated beamlines.

Methods and materials

Fred Hutch cancer center

FHCC is a three-room proton facility based on a Proteus Plus cyclotron (Ion Beam Applications, Louvain-La-Neuve, Belgium) in the Pacific Northwest of the USA. In the absence of a dedicated ocular beamline, a general-purpose fixed pencil beam scanning (PBS) beamline was optimized to achieve a sharp lateral penumbra [12], enabling the delivery of ocular treatments at our center. This center is among the few worldwide to treat ocular cases using PBS, integrating technologies such as MC calculations, inverse optimization, and beam shaping aperture systems. Treatment planning is performed with RayStation version 2023B (RaySearch Laboratories, Sweden) treatment planning system (TPS).

Study population

FHCC began offering PBS based ocular treatments starting in Jan of 2024. In one year, more than 50 patients were treated. For this report, a retrospective analysis was conducted on 40 patients with choroidal melanoma. We collected data on patient age, sex, tumor size as observed on brightness scan (b-scan) ultrasound, and tumor classification (stage). Additionally, we recorded any available symptomatic characteristics related to the tumor and the mode of disease discovery. We also noted the gross tumor volume (GTV) size and the minimum distance from critical organs at risk (OARs), such as the optic nerve, macula, and lacrimal gland. Finally, we examined the location of each patient's residence to determine how far they had to travel to receive treatment.

Ocular proton therapy workflow

The overall workflow for patients undergoing ocular treatments at FHCC proton therapy center is given in Fig. 1, with details provided in the following subsections.

Pre-simulation

Each patient underwent fundus photography and b-scan ultrasounds before tantalum marker placement. An ophthalmologist then evaluates the b-scan images to provide

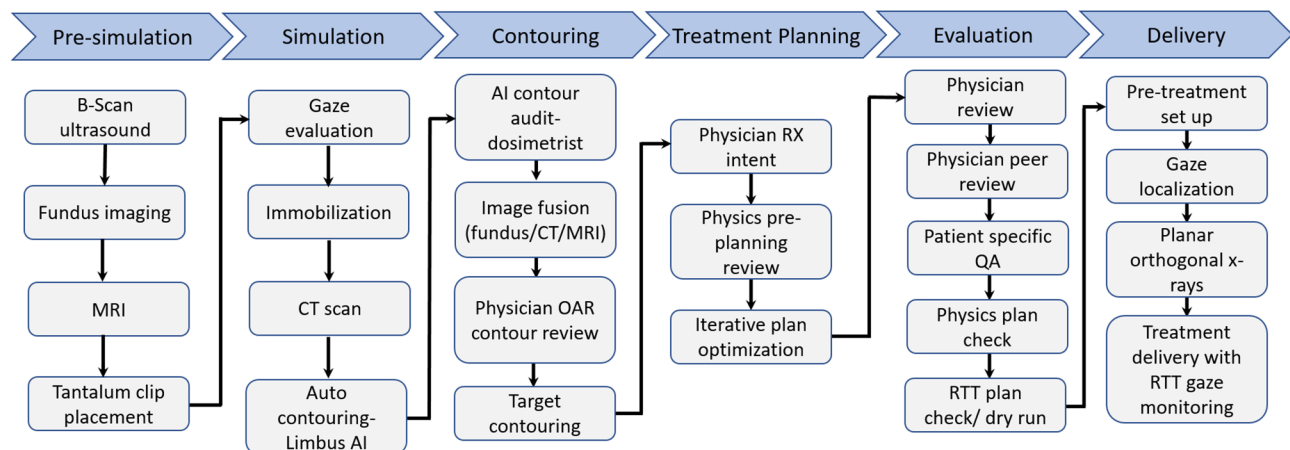


Fig. 1 The patient workflow for proton PBS-based ocular treatments, spanning from initial imaging to the final treatment

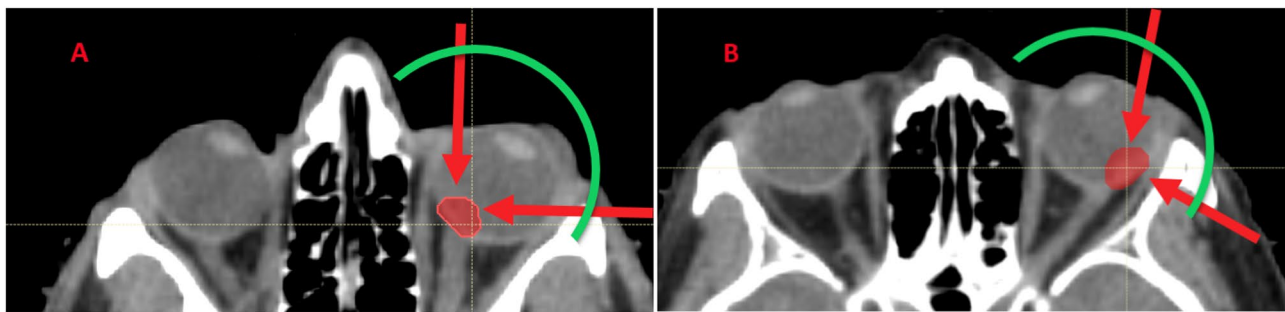


Fig. 2 An example of gaze directions for two different target locations is shown: (A) temporal gaze and (B) nasal gaze. The gaze angle is determined based on the MRI prior to the planning CT acquisition. Red arrows indicate the beam angles selected for treatment planning. Green arcs show the range of possible treatment beam angles

the radiation oncologist with the target's location, size, and extent. Fundus images are also passed on to radiation oncologist to aid target contouring.

Three tantalum markers are sutured into the treatment eye to enable daily image guidance using orthogonal X-rays, ensuring accurate reproduction of the gaze position established during CT simulation. Typically, these markers are placed after the MRI; however, in some cases, they may be placed beforehand. The placement is performed by the ophthalmologist following a standardized positioning protocol developed by our center and communicated to them in advance. This protocol intentionally places the markers away from the target area to minimize the likelihood of their interference with the treatment beam path. Unlike dedicated ocular systems that utilize marker positions for target definition, we use markers exclusively for gaze alignment.

The MRI images are obtained using a 3T Ingenia machine from Philips Healthcare. High-resolution MRI protocols are implemented to enable finer visualization of the optic structures. Three-dimensional (3-D) MRI images are acquired using the balanced fast field echo (bFFE) with SENSE technique, which provides high-resolution images for detailed examination, featuring a

slice thickness of 0.5 mm and an intra-slice resolution of 0.35 mm. An additional 3-D MRI scan is performed using the T2-weighted driven equilibrium radiofrequency reset pulse (DRIVE) sequence, achieving an isotropic resolution of 0.6 mm. Furthermore, T1-weighted axial scans with a thicker slice of 3 mm are obtained, both with and without the administration of gadolinium contrast agent, to assist with target delineation and contouring.

Simulation and immobilization

Before the CT scan, the attending radiation oncologist, medical physicist, and dosimetrist review the MRI scans of the patient's tumor to determine the optimal gaze angle and whether eyelid retractors are needed. There are generally three options for gaze: nasal, temporal, or neutral. The selected gaze angle chosen depends on the location of the tumor, the OARs nearby, and patient compliance. Examples of gaze angles are illustrated in Fig. 2. Patients can be treated either while lying supine or seated in a chair, with most being treated while seated due to the greater range of available treatment angles. When treated in a supine position, only lateral beam angles are possible at our center because treatments are delivered with a fixed horizontal beamline. Although potential beam

angles are evaluated before the CT scan, the decision regarding the treatment position—supine or seated—can be made during treatment planning. Other aspects of the CT simulation, such as immobilization, gaze localization, and the CT scan itself, are independent of the chosen treatment position. The eyelid retractor (SUH04, Storz Ophthalmic Instruments) is primarily used for patients who require treatment with a single anterior beam to keep the eyelid out of the beam path.

The chair (Fig. 3A) is an in-house manufactured device equipped with a kVue BOS insert (CQ Medical, Avondale, PA) as its backrest. The patient's head is immobilized using a Proton-B mask (Klarity Medical, Heath, OH), with a moldcare (Klarity Medical, Heath, OH) placed between the patient's head and the BOS frame. The backrest of the chair is fixed at a 20-degree recline to enhance patient comfort. The patient sits on a vaclock (Klarity Medical, Heath, OH), and the footrests can be adjusted to provide proper support for the patient's legs.

This treatment chair is integrated with IBA's robotic system, allowing for a full range of laterally coplanar beam angles. Due to the reclined position of the chair back, the treatment plane is pitched 20 degrees posteriorly. The integration enables patient imaging in the seated position, with the Adapt-Insight software (Ion Beam Applications, Belgium) automatically generating digitally reconstructed radiographs (DRRs) for various setup and treatment angles. During image guidance, 6-degree-of-freedom correction vectors can be automatically

transmitted to the robot to align the patient precisely. The chair supports patients weighing up to 285 pounds. While rare, limitations can arise for patients taller than 6 feet 4 inches, as the robot may reach its lower travel limit. In such cases, adjustments in patient positioning typically resolve the issue. Conversely, shorter patients may need a thicker Vac-Lok cushion to ensure proper positioning within the chair's upper travel range.

The in-house developed gaze immobilization system (Fig. 3B) is adapted from previously published work [16]. It features a rod-mounted camera that can be indexed both vertically and rotationally along a semicircular aluminum track attached to the BOS frame. The camera opening is marked with a small white dot, which the patient is instructed to focus on during both simulation and treatment to maintain gaze fixation. During the CT simulation process, we document and photograph the position of the rod on the track, the height of the camera on the rod, and the camera's rotation. These details are then uploaded to the MOSAIQ (Elekta Solutions AB, Sweden) record and verify system for daily verification prior to treatment.

Contouring and treatment planning

The treatment planning process begins with obtaining a head CT scan of the patient. This scan is performed using a wide-bore Somatom X.ceed (Siemens Healthineers, Germany). A helical scan with a slice thickness of 1 mm is acquired and reconstructed using a high-resolution

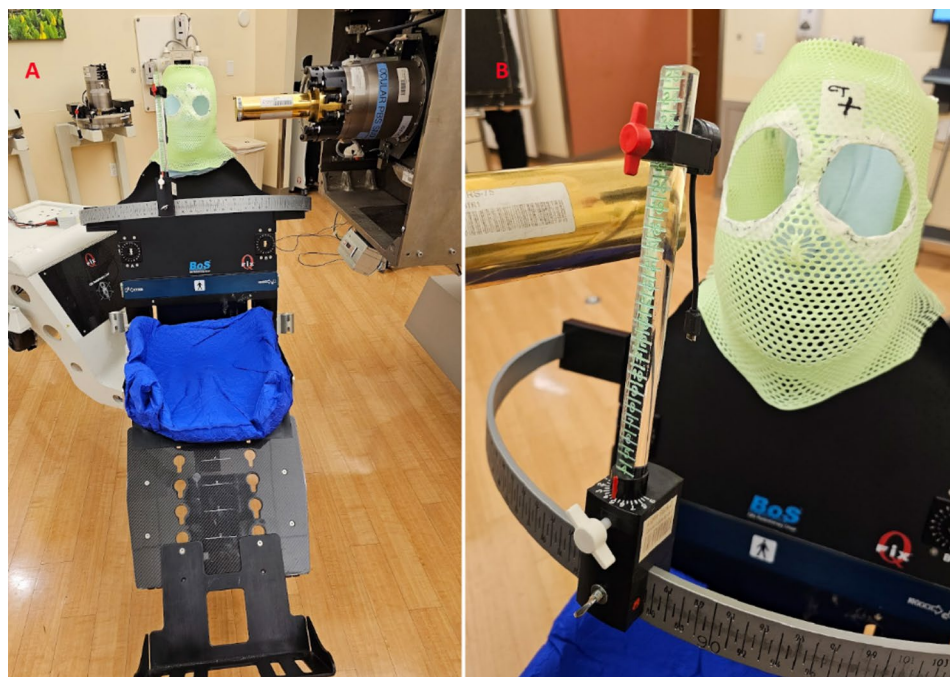


Fig. 3 (A) The typical patient setup includes a chair equipped with a Moldcare thermoplastic mask and a gaze localization and monitoring device. (B) The gaze localization and monitoring camera is mounted on indexed bars, allowing for adjustments in the superior-inferior direction and rotationally around the patient

40 cm field-of-view (FOV) for planning. This reconstruction is done both with and without iterative metal artifact correction (iMAR) algorithm. Additionally, an 85 cm FOV reconstruction is performed to identify the camera’s position, enabling the planner to exclude beam angles that might block its line of sight. The iMAR scan is useful for reducing streaking artifacts caused by the tantalum markers and is used for contouring the target and organs at risk (OARs). Meanwhile, dose calculations are performed on the conventionally reconstructed CT images. Artifacts caused by tantalum markers within the soft tissue or globe are overridden to unit (water) density for treatment planning. Artifacts in bony structures are overridden to bone-equivalent density, sampled from adjacent artifact-free regions. No override is applied directly to the tantalum markers themselves, as our placement strategy minimizes their presence in the beam path. In the uncommon instances where a beam intersects a marker—typically only one of the two beams—the markers are not overridden due to their small size and the limited resolution of CT, which prevents accurate contouring.

After acquiring the CT scan, relevant anatomical structures are automatically generated using Limbus AI (Radformation Inc., NY) and reviewed by a dosimetrist and a radiation oncologist. The optic disc is delineated as a 5 mm diameter, 2 mm height cylindrical volume positioned at the termination of the optic nerve within the sclera. Similarly, the macula is contoured as a 5 mm diameter, 2 mm height cylinder located laterally to the optic disc. The center-to-center distance between the optic disc and the macula is around 5 mm. A dosimetrist is responsible for fusing

the various CT and MRI images prior to target contouring. Since MRI is conducted with the patient’s gaze in a neutral position while the CT images are taken with the patient in a specific gaze, significant rotations are often necessary for proper CT-MRI image registration. RayOcular (RaySearch Laboratories, Sweden) [17], a commercial ocular treatment planning platform, has been implemented in our clinic—specifically its contouring module, which allows for fundus image registration with 3D CT/MRI scans. If requested, 2D fundus images can be registered to the volumetric datasets using the locations of the optic disc and macula, assuming a default center-to-center spacing of 4.2 mm. The radiation oncologist can then delineate the target using any of the three types of images: CT, MRI, or fundus. A 2 mm isotropic margin was added to the GTV to create the PTV, accounting for setup and target delineation uncertainties.

The treatment planning process is executed in RayStation using inverse planning techniques on the patient’s CT scan. Most treatments utilize two radiation beams, and typically, an additional 1.5–2 mm margin is included for aperture conformation around the PTV volume. The spots are arranged in a hexagonal pattern. Energy layer spacing is set to 0.2 times the width of the most distal Bragg peak at the 80% dose level, and spot spacing is set to 0.2 times the average projected sigma for each energy layer. Based on our empirical experience, this configuration provides sufficient spot and layer density for optimization while avoiding the upper monitor unit (MU) per spot limit imposed by our cyclotron-based system.

The dose-volume goals used for plan optimization are shown in Table 1. Treatments are given in five consecutive sessions, typically from Monday to Friday, with each session delivering a daily dose of 1000 cGy (RBE). Robust optimization using MC dose calculation algorithm is used to ensure that GTV coverage of 100% at 5000 cGy (RBE) dose for 3% range and 1 mm isotropic set-up perturbations. All the dose calculations are performed at grid size of 1 mm and final dose is calculated with statistical uncertainty of less than 0.5%. Air gap between the patient’s surface and the aperture are kept in the range of 3–5 cm to minimize lateral penumbra. The TPS is absolutely calibrated so that deliverable MUs can be obtained directly from the TPS without measurement or a separate calculation tool. While most treatments employ a two-beam approach, a single en face beam may be used in select cases where the tumor is shallow and located away from critical structures. When two beams are employed, each beam is optimized to deliver a uniform dose to the entire target volume.

Pre-treatment patient specific QA

The patient specific QA includes planar QA measurements using Gafchromic (Ashland Advanced Materials,

Table 1 Dose-Volume objectives used for treatment planning

Volume	Constraint	
	Dose [cGy(RBE)]	Volume (%)
GTV*	5000	100
PTV	5000	95
Optic Nerve	5000	max dose
Ciliary Body	2800	30
Lens	800	max dose
Lacrimal	2000	40
Retina	2200	44
Cornea+Limbus	5000	max dose
	3200	ave dose
	2500	58
	4500	32
Optic Disc	ALARA	
Macula	ALARA	
Brain	ALARA	

* GTV is robustly optimized to 1 mm isotropic set-up and 3% range perturbations

NJ) film and point dose measurements using a calibrated Pin-Point chamber 31,015 (PTW Freiburg GmbH). The treatment plan is mapped to a virtual water phantom and 1–2 depths are chosen for planar QA measurements. Due to the energy dependence of the film, the film analysis is performed in the relative mode, and a gamma index criterion of 3% dose and 1 mm distance-to-agreement is used. A gamma index of >90% for all points in the field where dose is greater than 10% of the maximum is considered acceptable. For point dose measurements, 1–2 depths on the central axis are selected, and differences between predicted and measured dose of less than 3% are considered acceptable. The in-room x-rays are used to verify the ion-chamber position within the field before measurements.

Set-up and treatment delivery

Patients are positioned daily using in-room orthogonal kilovoltage X-rays. The CT scan from the simulation session and the structure set from the TPS are imported into the Adapt-Insight software (Ion Beam Applications, Belgium) for image guidance. This software generates digitally reconstructed radiographs (DRRs) that are compared with the acquired X-rays to achieve image alignment.

The gaze angle is reproduced using indexed settings, and the patient is instructed to focus on a white dot at the tip of the camera. Initial X-rays are taken, and the patient is adjusted in six degrees of freedom to align the bony anatomy near the GTV. Once an acceptable bony alignment is achieved, the patient is asked to maintain gaze fixation, and a second set of X-rays is acquired to verify that tantalum markers are within 1 mm of their positions on the CT simulation images. For markers that are outside the 1 mm tolerance, patient's gaze is adjusted and verified on imaging again.

After confirming head alignment and gaze localization, treatment begins. The treatment delivery for a single beam takes 15 to 40 s. Following the completion of the first beam, the chair (or couch) is rotated to the next beam angle, and the X-ray alignment process is repeated. The total treatment time that consisted of the set-up, image-alignment, and treatment delivery for all the beams was calculated based on the patient's radio frequency identification (RFID) tag that is scanned when patient enters and exits the room. This time was adjusted to exclude any downtime caused by technical issues that prevented patient treatment, such as failures in proton beam delivery or malfunctions in the imaging system that may have occurred during treatment.

Case study

To illustrate the treatment planning process, a case study for a patient with choroidal melanoma of the left eye is

presented. The patient's isodose lines, clinical goals, and DVH statistics are presented.

Results

Study population

Table 2 summarizes the characteristics of the first 40 patients treated with PBS. The mean age of the patients was 63.9 years. Of the 40 patients, 18 were symptomatic, reporting symptoms such as difficulty focusing, blurry vision, dark spots, flashes, and floaters in their vision. The remaining 22 patients were asymptomatic and were diagnosed incidentally through various means, including routine eye exams, cataract surgery, diabetic eye exams, or diagnostic procedures for other conditions (e.g., a CT scan following a fall).

Out of 40 patients, 13 came from out of state for treatment, with three patients from Oregon, four from Montana, two from Alaska, two from Idaho, one from Hawaii, and one from Arizona. For Washington state patients, the average and median distance traveled to gain access to this treatment were 87 and 54 miles, respectively. For out-of-state patients, the average and median distance traveled were 933 and 605 miles, respectively.

Simulation and treatment planning

The apical height and largest basal diameter (LBD) measurements were obtained from ophthalmologists' evaluations based on b-scan ultrasound images, as detailed in Table 2. Figure 4 displays the axial CT views of the GTV and PTV for the initial 25 patients. Our institution treats ocular patients using either plaque brachytherapy or proton therapy. Plaque brachytherapy is typically offered to patients with melanomas in the anterior region. In contrast, proton therapy is primarily reserved for patients who are not suitable candidates for plaque brachytherapy, such as those with posteriorly situated lesions. The median GTV size, as determined from the treatment planning system, was 0.16 cc, with a range of 0.02 to 1.0 cc. Table 2 also presents the distances from the GTV to various OARs, including the lacrimal gland, macula, optic disc, and optic nerve. These measurements were taken from the planning CT and represent the distance from the edge of the GTV to the edge of the respective OAR. 22 out of 40 patients had GTV either touching or overlapping with macula or optic disc.

Of the 40 patients, 22 were simulated with a nasal gaze, 9 with a neutral gaze, and 9 with a lateral gaze. Only one patient out of 40 was simulated and treated using an eyelid retractor. For the remaining 39 patients, the use of an eyelid retractor was either deemed unnecessary or considered disadvantageous, as it could introduce skin folds into the beam path or restrict beam angles due to the retractor being within the treatment field. Five patients were treated in a supine position, while the remaining 35

Table 2 Clinical characteristics of first 40 choroidal melanoma patients treated with proton pencil beam scanning

Age at treatment (mean +/- SD) in years		63.9 +/- 13.6
Gender	Female	26
	Male	14
Eye	Left	21
	Right	19
	Symptomatic	18
	Asmptomatic	22
Tumor Characteristics		
Apical height (median, [range]) in mm		2.6 [0.7 , 11.9]
LBD (median, [range]) in mm		8.4 [2.8 , 15.5]
Tumor Classification		
cT1a		25
cT2a		9
cT2b		1
cT3a		4
cT4b		1
GTV Size (median, [range]) in cm ³		0.16 [0.02 , 0.89]
PTV Size (median, [range]) in cm ³		0.68 [0.17 , 2.27]
Minimum distance tumor edge to OAR (median, [range]) in mm		
Lacrimal Gland		5.1 [20 , 0]
Macula		0.6 [7 , 0]
Optic Disc		1.5 [10 , 0]
Optic Nerve		2.7 [11 , 0]

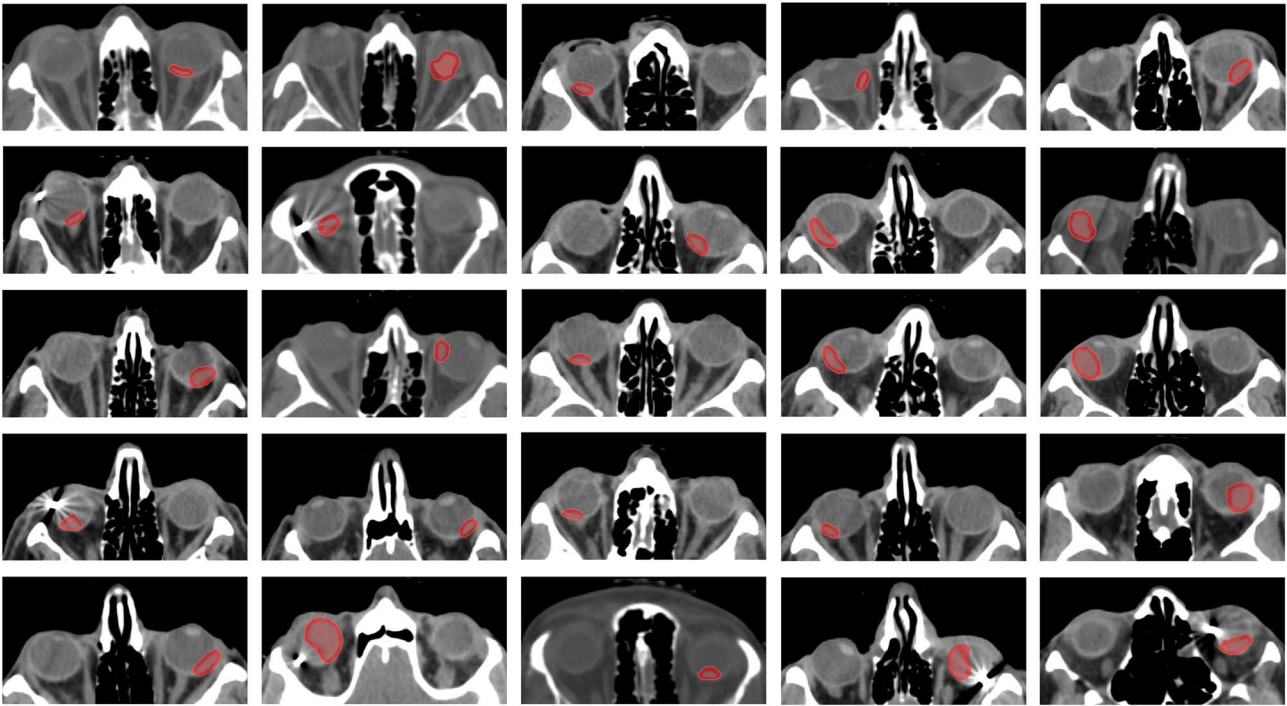


Fig. 4 CT cross-sections of the first 25 patients, displaying GTV contours in gray and PTV contours in red

were treated in a seated position using a chair. 34 patients were treated with two beams, while 6 patients were treated with a single beam.

Dose-volume statistics

The dose-volume statistics for the target and relevant OARs for the first 40 patients are presented in Fig. 5. All doses are reported as RBE-weighted doses, with an RBE factor of 1.1. The mean dose (Dmean) represents the average dose received by a specific OAR.

The D99% for the GTV ranged from 5041 to 5242 cGy (RBE), with a median value of 5075 cGy (RBE), while the V5000 cGy (RBE) for the PTV ranged from 95% to 99.8%, with a median of 95.9%. For optic structures, the median D2% dose was 4188 cGy (RBE) for the optic nerve, 5024 cGy (RBE) for the optic disc, and 5133 cGy (RBE) for the macula. Among the 40 patients, 22 had optic disc D2% >50 Gy (RBE), 8 had doses between 40 and 50 Gy (RBE), and 10 had doses <40 Gy (RBE); for the macula, 25 patients had D2% >50 Gy (RBE), 11 had 40–50 Gy (RBE), and 4 had <40 Gy (RBE); and for the optic nerve, 11 had D2% >50 Gy (RBE), 10 had 40–50 Gy (RBE), and 19 had <40 Gy (RBE). It should be noted that the macula and optic disc were modeled as cylindrical

structures with a diameter of 5 mm and a thickness of 2 mm, which are larger than their actual anatomical sizes and would therefore influence the reported dose metrics. The D2% dose to the ciliary body and lens was 1058 cGy (RBE) and 671 cGy (RBE), respectively. The lacrimal gland had a median Dmean dose of 1570 cGy (RBE) and a median D2% dose of 3850 cGy (RBE), with V2000 cGy (RBE) values ranging from 0% to 99.9% (median: 37%, mean: 35.5%). The retina received a median D2% dose of 5202 cGy (RBE) and a median Dmean dose of 1956 cGy (RBE), with a V2200 cGy (RBE) volume ranging from 18.6% to 63.9% (median: 39.6%). For the corneal limbus, the median Dmean dose was 30 cGy (RBE), and the median D2% dose was 44 cGy (RBE).

Sixteen out of 40 patients met all the clinical goals outlined in Table 1. Thirteen did not meet the lacrimal gland dose-volume constraint of V2000 cGy (RBE) <40%. For these 13 patients, the average and median distance between GTV and Lacrimal were 1.5 and 0.5 mm, respectively. For the same patient cohort, the average lacrimal volume receiving 2000 cGy (RBE) was 75.2%, with a range of 40.4% to 99.9%. Twelve patients did not meet the V2200 cGy (RBE) <44% retina dose-volume goal. For these patients, the average retina receiving 2200 cGy

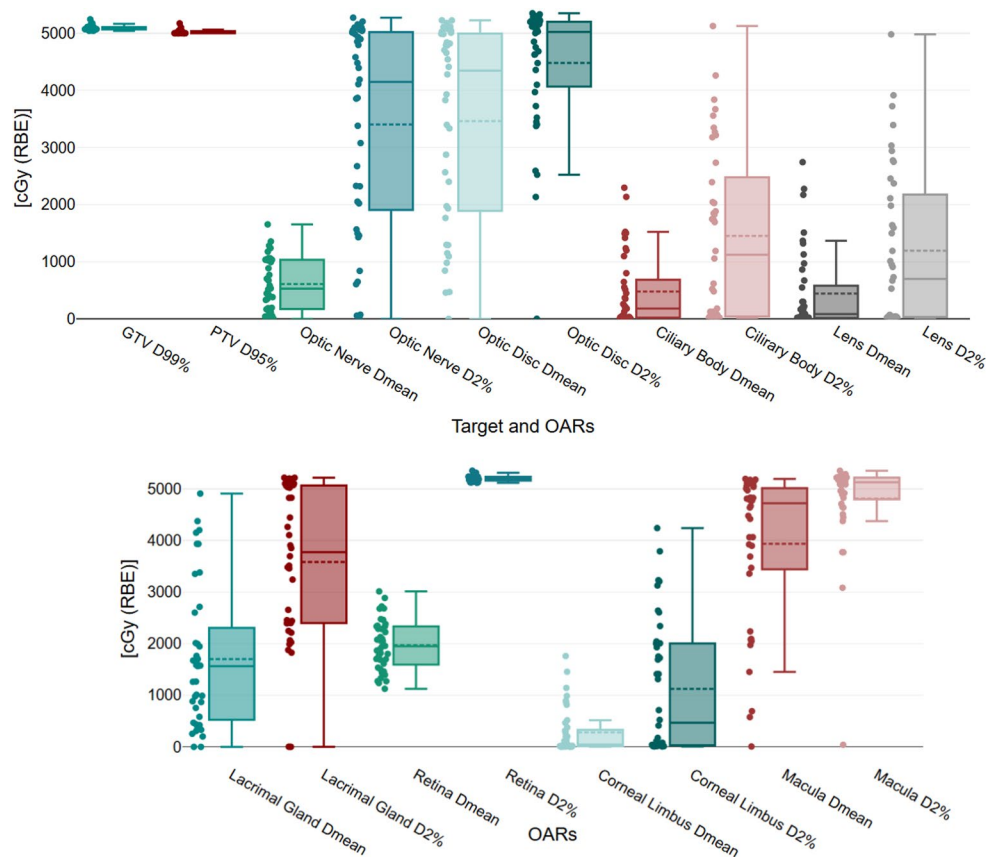


Fig. 5 Box plots illustrate dose-volume statistics for the first 40 patients, including targets and OARs. Whiskers represent the interquartile range (25–75%), while solid and dashed lines within the boxes indicate the median and mean values, respectively

(RBE) was 51.5%, ranging from 45.8 to 63.9%. Both the mean and median LBD for this subset of patients was 11.2 mm. These patients' mean and median GTV volumes were 0.5 and 0.4 cc, respectively. Ten patients did not meet the dose-volume constraint of max lens dose of 800 cGy (RBE). The average and median maximum lens dose for these patients was 2439 and 2178 cGy (RBE), respectively. All the patients met the corneal limbus goals, while only three patients did not meet the optic nerve max dose constraint of 5000 cGy (RBE). The average optic nerve max dose for these three patients was 5088 cGy (RBE).

Patient specific QA and total treatment time

Figure 6A and B present the results of the patient-specific QA. All 74 measurements fell within $\pm 3\%$ of the calculated values. For Gamma index analysis, all 74 planes achieved a Gamma index $> 90\%$, with mean and median values of 98.1% and 98.6%, respectively. Figure 6C illustrates the distribution of total treatment time spent in the room, as determined from RFID scans, with a mean treatment time of 25.4 min and a median of 24.3 min.

Case study

Figure 7 presents an example of a patient treated for choroidal melanoma using pencil beam scanning. The patient, a 64-year-old female, presents with a cT1a asymptomatic choroidal melanoma of the left eye that was discovered incidentally during a diabetic eye exam. The target had an apical height of 1.8 mm and an LBD of 7.2 mm, as noted on the b-scan ultrasound. The GTV was adjacent to the optic disc and macula, and within 1 mm of the optic nerve. For treatment planning, the patient was instructed to gaze nasally. The volumes of the GTV and PTV were 0.07 cc and 0.41 cc, respectively. Two beams—left lateral and left anterior oblique—were

used to uniformly cover the PTV (GTV plus a 2 mm isotropic margin) with a dose of 5000 cGy (RBE). All clinical goals outlined in Table 1 were achieved, with target coverage and OARs dose statistics detailed in Fig. 7.

Discussion

The classical model-based approach to ocular tumor treatment has been in use for over 40 years and has demonstrated high effectiveness in tumor control, eye retention, and minimizing complications. However, these dedicated systems are available only in select geographic regions, and limited access remains a significant barrier. Our approach, which utilizes non-dedicated systems for ocular treatment, necessitates the collection of comparable outcome data to evaluate and contrast our efficacy against that of dedicated systems.

To the best of our knowledge, no prior studies have reported the utilization of a general-purpose beamline for ocular proton therapy that primarily relies on commercially available tools while achieving beam characteristics comparable to those of specialized, dedicated beamlines. Several studies have investigated implementing ocular proton therapy using modern PBS systems. For example, Hickling et al. [18] developed an aperture system attached to a gantry designed to deliver ocular treatments with a sharp lateral penumbra. Their treatment planning process relies on Matlab scripts, employing one or five spots in the beam, depending on the required field size. However, their system depends on significant in-house development, which may not be feasible for many treatment centers. Moreover, while their approach is based on PBS delivery, the use of forward planning with one or five spots is intended to mimic the characteristics of scattering-based systems. In contrast, our methodology allows the inverse optimizer to design the treatment field without restrictions on the number of

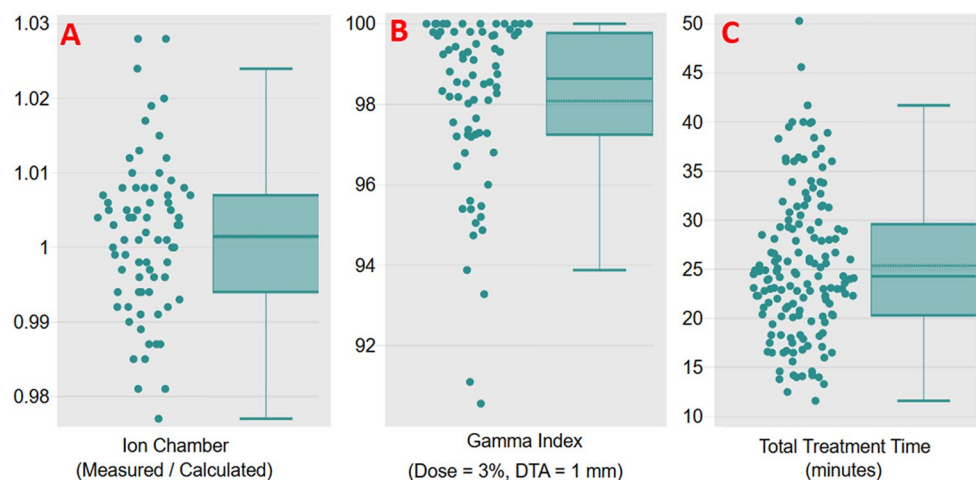


Fig. 6 (A) Agreement between measured and calculated point doses, (B) Gamma Index Analysis for all patient specific QA fields, (C) Distribution of total time spent by the patient inside the treatment room for each fraction

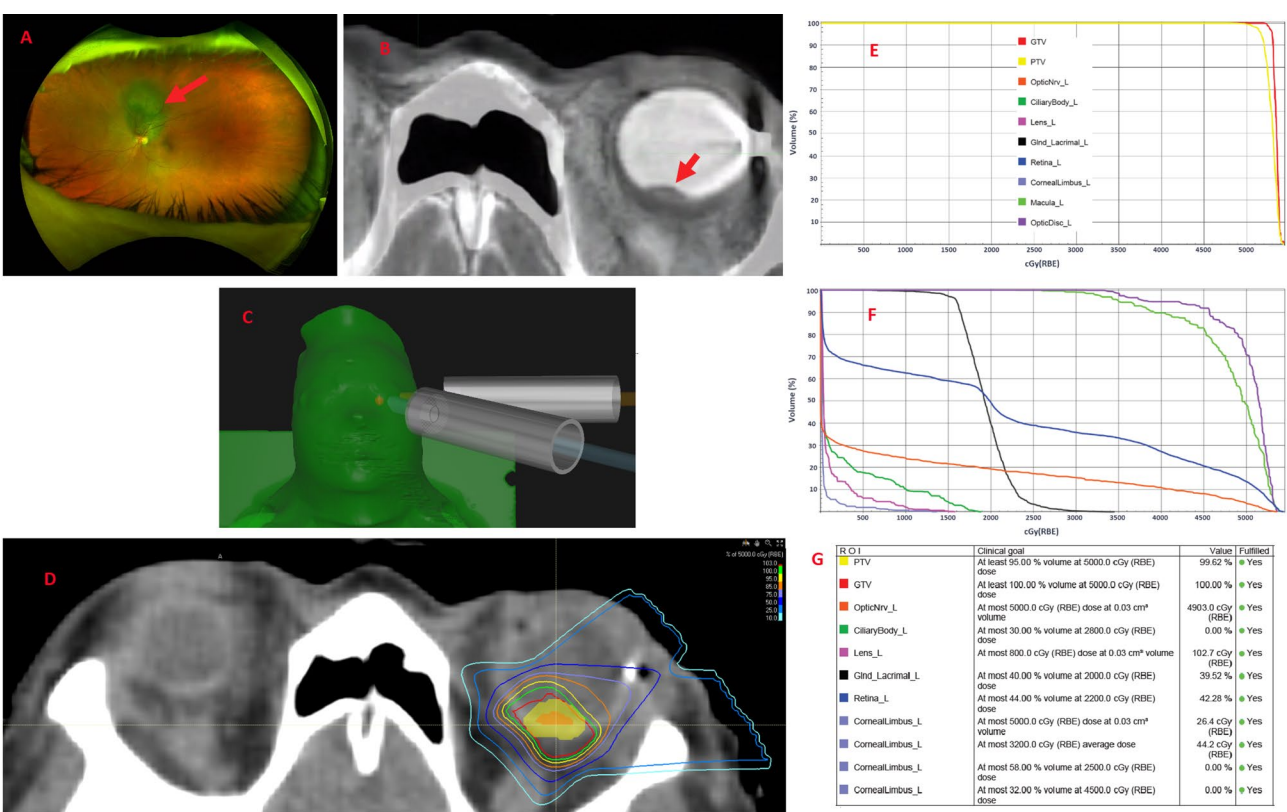


Fig. 7 An example patient case of a right-eye choroidal melanoma treated with pencil beam scanning is shown. **(A)** Fundus image from the ophthalmologist with a red arrow indicating the target. **(B)** MRI cross-sectional image showing the target location, also marked with a red arrow. **(C)** A two-beam approach was used to deliver a 5000 cGy (RBE) dose. **(D)** The isodose distribution on a planning CT cross-section displaying the GTV (red) and PTV (yellow) targets. **(E & F)** Dose-volume histograms for the target and relevant OARs. **(G)** A summary of the achieved clinical goals for the treatment plan

energy layers or spots, offering greater flexibility for dose uniformity inside the target.

Ciocca et al. [19] described their approach to adapting a synchrotron-based horizontal PBS beamline with an upstream range shifter for ocular treatments. Their system demonstrated beam properties suitable for ocular targets and reported treating nearly 100 patients within the first two years (2016–2018). Their method relies on research-based MC (FLUKA) for dose distribution and treatment planning, which may pose challenges for centers aiming to adopt similar programs. Our approach utilizes a commercially available RayStation TPS validated for small fields [12, 15], making it more accessible for widespread clinical implementation.

In this study, we presented dose-volume statistics for a cohort of 40 patients treated with PBS. While there are limited published reports providing similar data, comparisons between studies are complicated by several factors. A significant challenge is that most proton therapy planning for ocular melanomas relies on geometric models rather than CT-based data. Further complexities arise from the variability in treatment planning algorithms, ranging from basic lookup-based methods to more sophisticated approaches such as analytical or

MC algorithms. Differences in dose-fractionation regimens across clinics also add another layer of difficulty when comparing dose statistics. Hartsell et al. [16] published dosimetric data for 26 patients treated for ocular melanoma using a uniform scanning-based system with 3-dimensional treatment planning, albeit employing an analytical dose calculation algorithm and forward planning technique. They reported dose metrics for various OARs based on tumor location and staging. Their findings included a maximum point dose to the optic nerve ranging from 0 to 5200 cGy (RBE), which aligns closely with our observed range of 0 to 5272 cGy (RBE). Additionally, their study’s mean dose to the lacrimal gland ranged from 517 to 3022 cGy (RBE), compared to the interquartile range (Q1 to Q3) of 626 to 2454 cGy (RBE) in our data, as illustrated in Fig. 5.

Fleury et al. [20] evaluated one- and two-beam strategies for treating oculars using protons and provided dose-volume statistics for various OARs. They employed an in-house analytical dose calculation algorithm based on MRI images [21], which does not incorporate CT data for dose calculations. The authors reported a dosimetric benefit to the anterior segment and sclera when using a two-beam approach, although higher maximum doses

were observed for the macula and optic nerve. Box plots illustrating doses to the optic nerve, macula, and lacrimal gland for T1 and T2 tumors located within 3 mm of the optic nerve demonstrated a range of doses, from low levels (~ 10 cGy (RBE)) up to the prescription dose, which aligns with the findings presented in our Fig. 5. Although our data encompassed 0 to 11 mm distances between the optic nerve and the target, the mean and median distances were 3.2 mm and 2.7 mm, respectively.

Sharp lateral penumbras and steep distal fall-offs characterize most dedicated proton therapy facilities. Our modified system achieved lateral (80%- 20%) penumbras [12] from 1.4 to 2.5 mm depending on the depth, comparable to those of dedicated beamlines. However, the distal fall-off (90%-10%) in our system [12] ranges from 3.6 to 4.4 mm, depending on the beam energy. While these values are larger than some dedicated beamlines [11], where distal fall-offs of 1–2 mm have been reported, several other clinics [11] have reported distal fall-offs from 2.5 to 6.0 mm. Additionally, our current composite margin from the edge of the GTV to the aperture edge is 3.5–4 mm, which exceeds the 2.5 mm margins reported for dedicated systems [22]. As our program continues to mature and we gather more data on clinical outcomes and side effects, we aim to refine and optimize our margin strategy accordingly.

Wulff et al. [22] conducted a study simulating varying lateral and distal penumbras to create different treatment plans and evaluate doses of OARs. They found that plan quality is strongly influenced by the lateral penumbra and emphasized the importance of a sharp distal fall-off for posterior tumors near critical structures such as the macula, optic disc, and optic nerve. However, the findings of this study may not be directly applicable to our system for several reasons. First, their study was limited to beam angles primarily oriented anteriorly. Second, treatment planning was restricted to a single beam, as it relied on an eye model rather than the complete anatomical structure of the head, which allows for more flexibility in adding additional beams. Third, their planning was performed using a pencil beam algorithm rather than a MC based approach. Notably, Massachusetts General Hospital (MGH) [23–25] has successfully treated thousands of patients using a beamline with a distal fall-off of 6.0 mm, demonstrating that even with a less steep distal fall-off, effective treatment outcomes can be achieved.

Many future developments are possible that enhance the capabilities of our system. Current treatment planning is performed at isotropic resolution of 1 mm. RayStation 24B, that is in the process of FDA approval in the USA, would allow dose calculation at 0.5 mm resolution, bringing the capabilities of our system in line with dedicated beamline programs, most of which are capable of dose calculation resolution of 0.5 mm or better. As we

plan on CT, we are also able to utilize newer TPS features such as LET evaluation and incorporation of LET into the treatment planning. These new features could help for certain tumors where beam may be ending at the distal end, allowing higher LET regions to be placed inside the target and potentially sparing OARs of high LET areas.

Dual energy CTs are another possible technology that can lead to reduction in margins by reducing the range uncertainty. Current treatment planning is based on the indirect conversion to stopping power ratios inside the treatment planning system from element CT number. Due to this indirect conversion, higher uncertainty in the stopping can arise, as some tissues with the same CT number can have different stopping power. This error in SPR estimation is generally accounted by increasing the range robustness during the treatment planning. At our clinic, we use a range uncertainty of 3% for ocular targets. We are in the process of implementing a dual energy CT protocols that will yield direct SPR based on two different low and high energy CT scans. Some studies [26] have suggested that range uncertainty could be reduced to 1.7% from 3.5% based on the DECT based planning.

Further advancements to our gaze localization and tracking system are possible by introducing an automated software system that alerts the operator when the gaze deviates beyond the permitted tolerance. This would replace the current manual system, which depends on the operator's judgment to detect gaze deviations. We are also exploring improvements to beam quality that could be achieved by limiting energy slits to narrow the energy spectrum of protons in the beamline. While this may reduce the distal fall-off, it could also decrease the dose rate.

Conclusions

We have demonstrated that a robust ocular proton therapy program can be effectively implemented without requiring a dedicated beamline, primarily using commercially available tools and some in-house developments. A beam nozzle attachment incorporating a range shifter upstream—similar to our design [12]—can be easily fabricated for a high-energy general-purpose beamline, yielding beam qualities that allow treatment of ocular targets. At our facility, patients are treated while seated, using an in-house developed chair due to the limited availability of 360-degree gantry time. However, newer proton therapy centers with multiple gantries could implement an ocular proton therapy that allows for treatment on a traditional couch in a supine position. With ongoing advancements in TPS, including finer grid dose calculations and LET-based optimization, along with enhancements in CT imaging such as direct relative stopping power determination, PBS-based ocular proton therapy has the potential to provide benefits that are

not achievable with traditional dedicated systems. The follow-up data and outcomes for patients are being collected and will be reported in upcoming studies.

Abbreviations

GTV	Gross target volume
PTV	Planning target volume
OAR	Organ at risk
PBS	Pencil beam scanning
MC	Monte Carlo
TPS	Treatment planning system
DECT	Dual energy CT
CT	Computed Tomography
MRI	Magnetic Resonance Imaging
FOV	Field of view
RBE	Relative biological effectiveness
LET	Linear energy transfer
DVH	Dose Volume histogram
FHCC	Fred Hutchinson Cancer Center
QA	Quality assurance

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Author contributions

JS designed the project, gathered, analyzed, and interpreted the data and drafted the manuscript. AS, LH, RR, AE, CB, MS, RR and JC helped with manuscript revision. All authors helped design the techniques listed in the manuscript. All authors were part of the patient care outlined in the manuscript. All authors read and approved the final manuscript.

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Data availability

The datasets used and/or analyzed during the current study are available from the corresponding author on reasonable request.

Declarations

Ethics approval and consent to participate

All the patient analysis related work reported in this manuscript was approved under the University of Washington Institutional Review Board Study STUDY00010261.

Consent for publication

A consent for publication was obtained for the case report illustrated in this manuscript.

Competing interests

The authors declare no competing interests.

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